

2010.14265: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2010.14265_F00001	1	■ 1.000	$\{ \ }^{\{1,2\}}$	1,2	Alexander Marx ^{1,2}
2010.14265_F00002	1	■ 1.000	P	P	tribution P of the observed variables and try to infer the true
2010.14265_F00003	1	■ 1.000	G	G	causal graph G between them. Two standard assumptions
2010.14265_F00004	1	■ 0.999	X, Z	X, Z	graph (a) X, Z are fair independent coins and $Y := (X \oplus Z) \oplus E$,
2010.14265_F00005	1	■ 0.999	$Y := (X \oplus Z) \oplus E$	$Y := (X \oplus Z) \oplus E$	graph (a) X, Z are fair independent coins and $Y := (X \oplus Z) \oplus E$,
2010.14265_F00006	1	■ 0.999	X	X	noise term. Then X is independent of Y (denoted as $X \perp\!\!\!\perp_P Y$)
2010.14265_F00007	1	■ 0.999	Y	Y	noise term. Then X is independent of Y (denoted as $X \perp\!\!\!\perp_P Y$)
2010.14265_F00008	1	■ 0.999	$X \perp\!\!\!\perp_P Y$	$X \perp\!\!\!\perp_P Y$	noise term. Then X is independent of Y (denoted as $X \perp\!\!\!\perp_P Y$)
2010.14265_F00009	1	■ 1.000	$Z \perp\!\!\!\perp_P Y$	$Z \perp\!\!\!\perp_P Y$	and $Z \perp\!\!\!\perp_P Y$. Graph (b) could correspond with a linear model
2010.14265_F00010	1	■ 0.532	$X \perp\!\!\!\perp_P Z$	$X \perp\!\!\!\perp_P Z$	$X \perp\!\!\!\perp_P Y$, but $X \not\perp\!\!\!\perp_P Z$ and $Z \not\perp\!\!\!\perp_P Y$.
2010.14265_F00011	1	■ 0.532	$Z \not\perp\!\!\!\perp_P Y$	$Z \not\perp\!\!\!\perp_P Y$	$X \perp\!\!\!\perp_P Y$, but $X \not\perp\!\!\!\perp_P Z$ and $Z \not\perp\!\!\!\perp_P Y$.
2010.14265_F00012	1	■ 1.000	Z	Z	X and Z , that can be modelled by fair coins, together cause
2010.14265_F00013	1	■ 1.000	X, Y	X, Y	X, Y and Z in the causal graph—however, there are.
2010.14265_F00014	1	■ 1.000	37^{th}	37^{th}	Accepted for the 37 th Conference on Uncertainty in Artificial Intelligence (UAI 2021).
2010.14265_F00015	2	■ 1.000	$\boldsymbol{V}$	$\boldsymbol{V}$	random variables $\boldsymbol{V}$ with joint distribution P is defined such
2010.14265_F00016	2	■ 1.000	$X \rightarrow Y$	$X \rightarrow Y$	$X \rightarrow Y$ in G between two nodes representing the random
2010.14265_F00017	2	■ 1.000	$X, Y \in \boldsymbol{V}$	$X, Y \in \boldsymbol{V}$	variables $X, Y \in \boldsymbol{V}$ indicates that X is a <i>direct cause</i> or <i>parent</i>
2010.14265_F00018	2	■ 0.992	$X \in \boldsymbol{V}$	$X \in \boldsymbol{V}$	we denote the set of all parents of $X \in \boldsymbol{V}$ with $\text{Pa}(X)$, the set
2010.14265_F00019	2	■ 0.992	$\text{Pa}(X)$	$\text{Pa}(X)$	we denote the set of all parents of $X \in \boldsymbol{V}$ with $\text{Pa}(X)$, the set
2010.14265_F00020	2	■ 0.995	$\text{Ch}(X)$	$\text{Ch}(X)$	of all children with $\text{Ch}(X)$ and the set of parents and children
2010.14265_F00021	2	■ 0.294	$\text{PC}(X) := \text{Pa}(X) \cup \text{Ch}(X)$	$\text{PC}(X) := \text{Pa}(X) \cup \text{Ch}(X)$	with $\text{PC}(X) := \text{Pa}(X) \cup \text{Ch}(X)$. Further, we write $\text{An}(X)$ for
2010.14265_F00022	2	■ 0.294	$\text{An}(X)$	$\text{An}(X)$	with $\text{PC}(X) := \text{Pa}(X) \cup \text{Ch}(X)$. Further, we write $\text{An}(X)$ for
2010.14265_F00023	2	■ 0.999	$\text{De}(X)$	$\text{De}(X)$	the set of <i>ancestors</i> and denote its <i>descendants</i> by $\text{De}(X)$,
2010.14265_F00024	2	■ 0.999	$\text{Nd}(X) := \boldsymbol{V} \setminus \text{De}(X)$	$\text{Nd}(X) := \boldsymbol{V} \setminus \text{De}(X)$	we refer to the <i>non-descendants</i> of X as $\text{Nd}(X) := \boldsymbol{V} \setminus \text{De}(X)$.
2010.14265_F00025	2	■ 1.000	$\text{MB}(X) :=$	$\text{MB}(X) :=$	Last, the <i>Markov blanket</i> of X is defined as $\text{MB}(X) :=$
2010.14265_F00026	2	■ 0.617	$\text{PC}(X) \cup \text{Sp}(X)$	$\text{PC}(X) \cup \text{Sp}(X)$	$\text{PC}(X) \cup \text{Sp}(X)$, where $\text{Sp}(X)$ are the spouses of X , that
2010.14265_F00027	2	■ 0.617	$\text{Sp}(X)$	$\text{Sp}(X)$	$\text{PC}(X) \cup \text{Sp}(X)$, where $\text{Sp}(X)$ are the spouses of X , that
2010.14265_F00028	2	■ 1.000	d	d	d -separated of any other node in the graph given its Markov
2010.14265_F00029	2	■ 0.623	$\text{MB}(X)$	$\text{MB}(X)$	blanket and $\text{MB}(X)$ is the smallest such set.
2010.14265_F00030	2	■ 1.000	$Z \notin \boldsymbol{V}$	$Z \notin \boldsymbol{V}$	caused by an unobserved confounder $Z \notin \boldsymbol{V}$. This is also the
2010.14265_F00031	2	■ 0.783	$\mathcal{M} = (G, \boldsymbol{V}, P)$	$\mathcal{M} = (G, \boldsymbol{V}, P)$	defined above, we write $\mathcal{M} = (G, \boldsymbol{V}, P)$.
2010.14265_F00032	2	■ 1.000	$\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	$\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	Given three sets of random variables $\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$, where P
2010.14265_F00033	2	■ 1.000	$\boldsymbol{X}$	$\boldsymbol{X}$	is the joint distribution over $\boldsymbol{V}$, we denote that $\boldsymbol{X}$ is <i>proba-</i>
2010.14265_F00034	2	■ 1.000	$\boldsymbol{Y}$	$\boldsymbol{Y}$	<i>bilistically independent</i> of $\boldsymbol{Y}$ given $\boldsymbol{Z}$ in P as $\boldsymbol{X} \perp\!\!\!\perp_P \boldsymbol{Y} \mid \boldsymbol{Z}$.
2010.14265_F00035	2	■ 1.000	$\boldsymbol{Z}$	$\boldsymbol{Z}$	<i>bilistically independent</i> of $\boldsymbol{Y}$ given $\boldsymbol{Z}$ in P as $\boldsymbol{X} \perp\!\!\!\perp_P \boldsymbol{Y} \mid \boldsymbol{Z}$.
2010.14265_F00036	2	■ 1.000	$\boldsymbol{X} \perp\!\!\!\perp_P \boldsymbol{Y} \mid \boldsymbol{Z}$	$\boldsymbol{X} \perp\!\!\!\perp_P \boldsymbol{Y} \mid \boldsymbol{Z}$	<i>bilistically independent</i> of $\boldsymbol{Y}$ given $\boldsymbol{Z}$ in P as $\boldsymbol{X} \perp\!\!\!\perp_P \boldsymbol{Y} \mid \boldsymbol{Z}$.
2010.14265_F00037	2	■ 1.000	D	D	<i>D-separation</i> (Pearl, 2009) is defined in terms of paths. A
2010.14265_F00038	2	■ 0.998	p	p	<i>path</i> p between X and Y , denoted $p = \langle X, \dots, Y \rangle$, is a se-
2010.14265_F00039	2	■ 0.998	$p = \langle X, \dots, Y \rangle$	$p = \langle X, \dots, Y \rangle$	<i>path</i> p between X and Y , denoted $p = \langle X, \dots, Y \rangle$, is a se-
2010.14265_F00040	2	■ 1.000	$X_1, \dots, X_n$	$X_1, \dots, X_n$	quence of distinct nodes $X_1, \dots, X_n$ such that X_i is adjacent
2010.14265_F00041	2	■ 1.000	X_i	X_i	quence of distinct nodes $X_1, \dots, X_n$ such that X_i is adjacent

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2010.14265_F00042	2	■ 1.000	X_{i+1}	X_{i+1}	to X_{i+1} for $i = 1, \dots, n-1$, $X_1 = X$ and $X_n = Y$. Further, we
2010.14265_F00043	2	■ 1.000	$i=1, \ldots, n-1, X_1=X$	$i = 1, \dots, n-1, X_1 = X$	to X_{i+1} for $i = 1, \dots, n-1$, $X_1 = X$ and $X_n = Y$. Further, we
2010.14265_F00044	2	■ 1.000	$X_{\{n\}}=Y$	$X_n = Y$	to X_{i+1} for $i = 1, \dots, n-1$, $X_1 = X$ and $X_n = Y$. Further, we
2010.14265_F00045	2	■ 1.000	C	C	call a node C a <i>collider</i> on a path $\langle \dots, X, C, Y, \dots \rangle$, where
2010.14265_F00046	2	■ 1.000	$\langle \ldots, X, C, Y, \ldots \rangle$	$\langle \dots, X, C, Y, \dots \rangle$	call a node C a <i>collider</i> on a path $\langle \dots, X, C, Y, \dots \rangle$, where
2010.14265_F00047	2	■ 1.000	$X \rightarrow C \leftarrow Y$	$X \rightarrow C \leftarrow Y$	that is $X \rightarrow C \leftarrow Y$.
2010.14265_F00048	2	■ 0.636	$\mathbf{Z}$	$\mathbf{Z}$	1. every non-collider on the path is not in $\mathbf{Z}$, and
2010.14265_F00049	2	■ 0.957	$X \in \mathbf{X}$	$X \in \mathbf{X}$	given $\mathbf{Z}$, if for every pair X, Y , with $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$, X
2010.14265_F00050	2	■ 0.957	$Y \in \mathbf{Y}, X$	$Y \in \mathbf{Y}, X$	given $\mathbf{Z}$, if for every pair X, Y , with $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$, X
2010.14265_F00051	2	■ 1.000	$\mathbf{X} \perp_G \mathbf{Y} \mid \mathbf{Z}$	$\mathbf{X} \perp_G \mathbf{Y} \mid \mathbf{Z}$	$\mathbf{X} \perp_G \mathbf{Y} \mid \mathbf{Z}$ if X is d -separated from Y given $\mathbf{Z}$. A useful
2010.14265_F00052	3	■ 0.539	$\mathcal{M} =$	$\mathcal{M} =$	Definition 3 (Adjacency Faithfulness) Let $\mathcal{M} =$
2010.14265_F00053	3	■ 1.000	$(G, \mathbf{V}, P)$	$(G, \mathbf{V}, P)$	$(G, \mathbf{V}, P)$, if $X, Y \in \mathbf{V}$ are adjacent in G , then they
2010.14265_F00054	3	■ 1.000	$\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Y\}$	$\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Y\}$	are probabilistically dependent given all $\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Y\}$.
2010.14265_F00055	3	■ 0.745	$G, \mathbf{V}, P$	$G, \mathbf{V}, P$	$(G, \mathbf{V}, P)$ and let $\langle X, Y, Z \rangle$ be an unshielded path ¹ in G .
2010.14265_F00056	3	■ 0.745	$\langle X, Y, Z \rangle$	$\langle X, Y, Z \rangle$	$(G, \mathbf{V}, P)$ and let $\langle X, Y, Z \rangle$ be an unshielded path ¹ in G .
2010.14265_F00057	3	■ 1.000	$X \rightarrow Y \leftarrow Z$	$X \rightarrow Y \leftarrow Z$	1. If $X \rightarrow Y \leftarrow Z$, then X and Z are dependent given any
2010.14265_F00058	3	■ 0.562	$V \setminus \{X, Z\}$	$V \setminus \{X, Z\}$	subset in $V \setminus \{X, Z\}$ that contains Y ; otherwise
2010.14265_F00059	3	■ 0.990	$X, Y, Z \in \mathbf{V}$	$X, Y, Z \in \mathbf{V}$	a triple of nodes $X, Y, Z \in \mathbf{V}$ s.t. $X \rightarrow Y \leftarrow Z$, where at least
2010.14265_F00060	3	■ 1.000	$\langle X, Y, Z \rangle, X$	$\langle X, Y, Z \rangle, X$	¹ For an unshielded path $\langle X, Y, Z \rangle$, X is adjacent to Y and Y is
2010.14265_F00061	3	■ 1.000	$X, Z \sim N(0, 1)$	$X, Z \sim N(0, 1)$	where $X, Z \sim N(0, 1)$ are iid and $Y := \text{sign}(XZ) \cdot E$, with $E \sim$
2010.14265_F00062	3	■ 1.000	$Y := \text{sign}(XZ) \cdot E$	$Y := \text{sign}(XZ) \cdot E$	where $X, Z \sim N(0, 1)$ are iid and $Y := \text{sign}(XZ) \cdot E$, with $E \sim$
2010.14265_F00063	3	■ 1.000	$E \sim$	$E \sim$	where $X, Z \sim N(0, 1)$ are iid and $Y := \text{sign}(XZ) \cdot E$, with $E \sim$
2010.14265_F00064	3	■ 1.000	$\text{Exp}\left(\frac{1}{\sqrt{2}}\right)$	$\text{Exp}\left(\frac{1}{\sqrt{2}}\right)$	$\text{Exp}(\frac{1}{\sqrt{2}})$. The dependence is only detectable by considering
2010.14265_F00065	3	■ 0.992	$\mathcal{M} = (G, \mathbf{V}, P)$	$\mathcal{M} = (G, \mathbf{V}, P)$	Example 1 Let $\mathcal{M} = (G, \mathbf{V}, P)$ be a causal model. Given
2010.14265_F00066	3	■ 0.890	$X \oplus Z :=$	$X \oplus Z :=$	as $Y := (X \oplus Z) \oplus E$, where $\oplus$ denotes xor—i.e. $X \oplus Z :=$
2010.14265_F00067	3	■ 0.699	$(X+Z) \bmod 2$	$(X + Z) \bmod 2$	$(X + Z) \bmod 2$ —and E is a biased coin with $P(E = 1) = p$,
2010.14265_F00068	3	■ 0.699	E	E	$(X + Z) \bmod 2$ —and E is a biased coin with $P(E = 1) = p$,
2010.14265_F00069	3	■ 0.699	$P(E=1)=p$	$P(E = 1) = p$	$(X + Z) \bmod 2$ —and E is a biased coin with $P(E = 1) = p$,
2010.14265_F00070	3	■ 0.693	$0 \leq p < \frac{1}{2}$	$0 \leq p < \frac{1}{2}$	where $0 \leq p < \frac{1}{2}$ and $E \perp_P \{X, Z\}$. Hence, $X \not\perp_G Y$, $Z \not\perp_G Y$,
2010.14265_F00071	3	■ 0.693	$E \perp_P \{X, Z\}$	$E \perp_P \{X, Z\}$	where $0 \leq p < \frac{1}{2}$ and $E \perp_P \{X, Z\}$. Hence, $X \not\perp_G Y$, $Z \not\perp_G Y$,
2010.14265_F00072	3	■ 0.693	$X \perp_G Y, Z \perp_G Y$	$X \perp_G Y, Z \perp_G Y$	where $0 \leq p < \frac{1}{2}$ and $E \perp_P \{X, Z\}$. Hence, $X \not\perp_G Y$, $Z \not\perp_G Y$,
2010.14265_F00073	3	■ 0.949	$Y \not\perp_P \{X, Z\}$	$Y \not\perp_P \{X, Z\}$	check the joint distribution, we can find that $Y \not\perp_P \{X, Z\}$, or
2010.14265_F00074	3	■ 0.998	$X \perp_P Z \mid Y$	$X \perp_P Z \mid Y$	$X \not\perp_P Z \mid Y$, since we get that $P(X = 1, Z = 1, Y = 1) = \frac{p}{4}$,
2010.14265_F00075	3	■ 0.998	$P(X=1, Z=1, Y=1)=\frac{p}{4}$	$P(X = 1, Z = 1, Y = 1) = \frac{p}{4}$	$X \not\perp_P Z \mid Y$, since we get that $P(X = 1, Z = 1, Y = 1) = \frac{p}{4}$,
2010.14265_F00076	3	■ 1.000	$P(X=1, Z=1) \cdot P(Y=1)=\frac{1}{4} \cdot \frac{1}{2}=\frac{1}{8}$	$P(X = 1, Z = 1) \cdot P(Y = 1) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$	where $P(X = 1, Z = 1) \cdot P(Y = 1) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$. Those terms
2010.14265_F00077	3	■ 1.000	$p=\frac{1}{2}$	$p = \frac{1}{2}$	are only equal if $p = \frac{1}{2}$, which we excluded by assumption.
2010.14265_F00078	3	■ 1.000	$E \sim \text{Exp}\left(\frac{1}{\sqrt{2}}\right)$	$E \sim \text{Exp}\left(\frac{1}{\sqrt{2}}\right)$	$E \sim \text{Exp}(\frac{1}{\sqrt{2}})$ (see Figure 2). The authors demonstrate that
2010.14265_F00079	4	■ 0.914	(X)	(X)	pills (X) can influence the risk of getting thrombosis (Y)
2010.14265_F00080	4	■ 0.914	(Y)	(Y)	pills (X) can influence the risk of getting thrombosis (Y)
2010.14265_F00081	4	■ 1.000	$Z := \alpha X, Y := \beta Z - \gamma X$	$Z := \alpha X, Y := \beta Z - \gamma X$	system in which $Z := \alpha X$, $Y := \beta Z - \gamma X$ and $\gamma = \alpha\beta$.
2010.14265_F00082	4	■ 1.000	$\gamma = \alpha\beta$	$\gamma = \alpha\beta$	system in which $Z := \alpha X$, $Y := \beta Z - \gamma X$ and $\gamma = \alpha\beta$.
2010.14265_F00083	4	■ 1.000	$X \rightarrow Z \leftarrow Y$	$X \rightarrow Z \leftarrow Y$	$X \rightarrow Z \leftarrow Y$ is also a valid graph for those independencies—
2010.14265_F00084	4	■ 1.000	$Z \rightarrow W \rightarrow Y$	$Z \rightarrow W \rightarrow Y$	there is a path $Z \rightarrow W \rightarrow Y$ (Zhang and Spirtes, 2008).
2010.14265_F00085	4	■ 1.000	$G^{\prime}$	$G^{\prime}$	to be more frugal than $G^{\prime}$, if G contains fewer edges than
2010.14265_F00086	4	■ 0.995	$\{X, Y, Z\}$	$\{X, Y, Z\}$	we call $\{X, Y, Z\}$ an <i>unfaithful triple</i> w.r.t. P^3 . If further for
2010.14265_F00087	4	■ 0.995	P^3	P^3	we call $\{X, Y, Z\}$ an <i>unfaithful triple</i> w.r.t. P^3 . If further for

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2010.14265_F00088	4	■ 1.000	$A, B \in \{X, Y, Z\}$	$A, B \in \{X, Y, Z\}$	<i>each distinct pair of nodes $A, B \in \{X, Y, Z\}$:</i>
2010.14265_F00089	4	■ 0.900	$X \nvdash_P \{Y, Z\}$	$X \not\vdash_P \{Y, Z\}$	³ Not mutually independent implies that $X \not\vdash_P \{Y, Z\}$,
2010.14265_F00090	4	■ 0.550	$Y \nvdash_P \{X, Z\}$	$Y \not\vdash_P \{X, Z\}$	$Y \not\vdash_P \{X, Z\}$ or $Z \not\vdash_P \{X, Y\}$.
2010.14265_F00091	4	■ 0.550	$Z \not\measuredangle_P \{X, Y\}$	$Z \not\lhd_P \{X, Y\}$	$Y \not\vdash_P \{X, Z\}$ or $Z \not\vdash_P \{X, Y\}$.
2010.14265_F00092	5	■ 1.000	U	U	Since X is d -separated from Y given U and Z , they do
2010.14265_F00093	5	■ 1.000	$\{U, Y, Z\}$	$\{U, Y, Z\}$	not form a minimal unfaithful triple. Neither do $\{U, Y, Z\}$,
2010.14265_F00094	5	■ 1.000	$\{W, Z\}$	$\{W, Z\}$	since Y can be d -separated from U given $\{W, Z\}$. Thus, only
2010.14265_F00095	5	■ 0.925	$\{U, W, Z\}$	$\{U, W, Z\}$	$\{U, W, Z\}$ can be a minimal unfaithful triple.
2010.14265_F00096	5	■ 0.243	$X \nVdash_P \{Y, Z\}$	$X \not\vdash_P \{Y, Z\}$	tween all three variables, e.g. $X \not\vdash_P \{Y, Z\}$, implies that also
2010.14265_F00097	5	■ 0.959	$Y \nVdash_P \{X, Z\}$	$Y \not\vdash_P \{X, Z\}$	$Y \not\vdash_P \{X, Z\}$ and $Z \not\vdash_P \{X, Y\}$.
2010.14265_F00098	5	■ 0.959	$Z \not\perp_P \{X, Y\}$	$Z \not\perp_P \{X, Y\}$	$Y \not\vdash_P \{X, Z\}$ and $Z \not\vdash_P \{X, Y\}$.
2010.14265_F00099	5	■ 0.803	$\{X, Y, Z\} \subseteq \boldsymbol{V}$	$\{X, Y, Z\} \subseteq \boldsymbol{V}$	Lemma 1 <i>Given $\mathcal{M} = (G, \boldsymbol{V}, P)$, let $\{X, Y, Z\} \subseteq \boldsymbol{V}$ form</i>
2010.14265_F00100	5	■ 0.539	$X \perp_P \{Y, Z\}, Y \not\perp_P \{X, Z\}$	$X \perp_P \{Y, Z\}, Y \not\perp_P \{X, Z\}$	<i>an unfaithful triple in P, then $X \not\vdash_P \{Y, Z\}$, $Y \not\vdash_P \{X, Z\}$ and</i>
2010.14265_F00101	5	■ 0.956	$Z \perp_P \{X, Y\}$	$Z \perp_P \{X, Y\}$	<i>$Z \not\vdash_P \{X, Y\}$, which in addition implies that $X \not\vdash_P Y \mid Z$,</i>
2010.14265_F00102	5	■ 0.956	$X \perp_P Y \mid Z$	$X \perp_P Y \mid Z$	<i>$Z \not\vdash_P \{X, Y\}$, which in addition implies that $X \not\vdash_P Y \mid Z$,</i>
2010.14265_F00103	5	■ 0.484	$Y, \measuredangle_P Z \mid X$	$Y, \angle_P Z \mid X$	<i>$X \not\vdash_P Z \mid Y$ and $Y \not\vdash_P Z \mid X$.</i>
2010.14265_F00104	5	■ 1.000	X, Y, Z	X, Y, Z	Consider Example 1. Since X, Y, Z form an unfaithful triple,
2010.14265_F00105	5	■ 0.999	$C \ M \ C$	CMC	<i>unfaithful triple in P. If CMC holds, each node in the triple</i>
2010.14265_F00106	5	■ 0.422	$X, Y, Z \in V$	$X, Y, Z \in V$	Theorem 1 <i>Given $\mathcal{M} = (G, \boldsymbol{V}, P)$ with distinct $X, Y, Z \in V$</i>
2010.14265_F00107	5	■ 0.960	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y, Z\}$	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y, Z\}$	<i>and assume that CMC holds. If $\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y, Z\}$ it holds</i>
2010.14265_F00108	5	■ 0.198	$X, \not\perp_P Y \mid Z \cup S, X, \perp_P Z \mid Y \cup S$	$X, \not\perp_P Y \mid Z \cup S, X, \perp_P Z \mid Y \cup S$	<i>that $X \not\vdash_P Y \mid Z \cup S$, $X \not\vdash_P Z \mid Y \cup S$ and $Y \not\vdash_P Z \mid X \cup S$,</i>
2010.14265_F00109	5	■ 0.198	$Y, \perp_P Z \mid X \cup S$	$Y, \perp_P Z \mid X \cup S$	<i>that $X \not\vdash_P Y \mid Z \cup S$, $X \not\vdash_P Z \mid Y \cup S$ and $Y \not\vdash_P Z \mid X \cup S$,</i>
2010.14265_F00110	5	■ 1.000	k	k	Definition 6 (k-Association) <i>Let P be the joint distribution</i>
2010.14265_F00111	5	■ 0.448	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y\} : X \not\vdash_P Y \mid \boldsymbol{S}$	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y\} : X \not\vdash_P Y \mid \boldsymbol{S}$	<i>to Y, if $\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y\} : X \not\vdash_P Y \mid \boldsymbol{S}$.</i>
2010.14265_F00112	5	■ 0.943	$X, Y_{\{1\}}, Y_{\{2\}} \in \boldsymbol{V}, X$	$X, Y_1, Y_2 \in \boldsymbol{V}, X$	<i>2. Given distinct $X, Y_1, Y_2 \in \boldsymbol{V}$, X is 2-associated to</i>
2010.14265_F00113	5	■ 1.000	$\left\{Y_{\{1\}}, Y_{\{2\}}\right\}$	$\{Y_1, Y_2\}$	<i>$\{Y_1, Y_2\}$ if $\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y_1, Y_2\}$ it holds that</i>
2010.14265_F00114	5	■ 1.000	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y_{\{1\}}, Y_{\{2\}}\}$	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y_1, Y_2\}$	<i>$\{Y_1, Y_2\}$ if $\forall \boldsymbol{S} \subseteq \boldsymbol{V} \setminus \{X, Y_1, Y_2\}$ it holds that</i>
2010.14265_F00115	5	■ 0.612	$X \nvdash_P Y_{\{1\}} \mid \boldsymbol{S} \cup Y_{\{2\}}$	$X \not\vdash_P Y_1 \mid \boldsymbol{S} \cup Y_2$	<i>i) $X \not\vdash_P Y_1 \mid \boldsymbol{S} \cup Y_2$,</i>
2010.14265_F00116	5	■ 0.446	$X \nvdash_P Y_{\{2\}} \mid \boldsymbol{S} \cup Y_{\{1\}}$	$X \not\vdash_P Y_2 \mid \boldsymbol{S} \cup Y_1$	<i>ii) $X \not\vdash_P Y_2 \mid \boldsymbol{S} \cup Y_1$ and</i>
2010.14265_F00117	5	■ 0.702	$Y_{\{1\}} \not\perp_P Y_{\{2\}} \mid \boldsymbol{S} \cup X$	$Y_1 \not\perp_P Y_2 \mid \boldsymbol{S} \cup X$	<i>iii) $Y_1 \not\vdash_P Y_2 \mid \boldsymbol{S} \cup X$.</i>
2010.14265_F00118	5	■ 0.958	$Y_{\{1\}}$	Y_1	<i>associated to $\{Y_1, Y_2\}$ and not 1-associated to Y_1 or Y_2.</i>
2010.14265_F00119	5	■ 0.958	$Y_{\{2\}}$	Y_2	<i>associated to $\{Y_1, Y_2\}$ and not 1-associated to Y_1 or Y_2.</i>
2010.14265_F00120	5	■ 0.863	$X_{\{2\}} \setminus \{Y, Z\}$	$X_{-2} \{Y, Z\}$	and write $X_{-2} \{Y, Z\}$ if X is 2-associated to Y and Z resp.
2010.14265_F00121	5	■ 0.572	$X_{\{ \}_{1}} Y$	$X_{-1} Y$	$X_{-1} Y$ if X is 1-associated to Y . We denote a strict 2-
2010.14265_F00122	5	■ 0.906	$\{ \}_{2}$	2	association by “ $\overset{s}{-}_2$ ”. If we refer to a set $\boldsymbol{Y}$ that contains
2010.14265_F00123	5	■ 0.971	$X_{\leq 2} \boldsymbol{Y}$	$X_{\leq 2} \boldsymbol{Y}$	1- or 2-associated to this set, we write $X_{\leq 2} \boldsymbol{Y}$. Similarly,
2010.14265_F00124	5	■ 0.888	$X \stackrel{s}{\leq 2} \boldsymbol{Y}$	$X \stackrel{s}{\leq 2} \boldsymbol{Y}$	we write $X \stackrel{s}{\leq 2} \boldsymbol{Y}$, if X is 1- or strictly 2-associated to $\boldsymbol{Y}$.
2010.14265_F00125	5	■ 1.000	$X \rightarrow Y \rightarrow Z$	$X \rightarrow Y \rightarrow Z$	chain $X \rightarrow Y \rightarrow Z$, where no adjacency failure occurs. In
2010.14265_F00126	5	■ 0.758	$X_{-1} Y$	$X_{-1} Y$	this case, $X_{-1} Y$ and $Y_{-1} Z$. Triple interactions that match
2010.14265_F00127	5	■ 0.758	$Y_{-1} Z$	$Y_{-1} Z$	this case, $X_{-1} Y$ and $Y_{-1} Z$. Triple interactions that match
2010.14265_F00128	5	■ 0.970	$X \stackrel{s}{\rightarrow} \{Y, Z\}, Y \stackrel{s}{\rightarrow} \{X, Z\}$	$X \stackrel{s}{\rightarrow} \{Y, Z\}, Y \stackrel{s}{\rightarrow} \{X, Z\}$	matches the definition, since $X \stackrel{s}{\rightarrow} \{Y, Z\}$, $Y \stackrel{s}{\rightarrow} \{X, Z\}$ and
2010.14265_F00129	6	■ 0.681	$Z \stackrel{s}{\rightarrow} \{X, Y\}$	$Z \stackrel{s}{\rightarrow} \{X, Y\}$	$Z \stackrel{s}{\rightarrow} \{X, Y\}$. In general, strict 2-associations describe col-
2010.14265_F00130	6	■ 0.744	$\boldsymbol{Y} \subseteq MB(X)$	$\boldsymbol{Y} \subseteq MB(X)$	<i>in G, there exists $\boldsymbol{Y} \subseteq MB(X)$, with $Y \in \boldsymbol{Y}$, s.t. $X \stackrel{s}{\leq 2} \boldsymbol{Y}$.</i>
2010.14265_F00131	6	■ 0.744	$Y \in \boldsymbol{Y}$	$Y \in \boldsymbol{Y}$	<i>in G, there exists $\boldsymbol{Y} \subseteq MB(X)$, with $Y \in \boldsymbol{Y}$, s.t. $X \stackrel{s}{\leq 2} \boldsymbol{Y}$.</i>
2010.14265_F00132	6	■ 0.744	$X \stackrel{s}{\leq 2} \boldsymbol{Y}$	$X \stackrel{s}{\leq 2} \boldsymbol{Y}$	<i>in G, there exists $\boldsymbol{Y} \subseteq MB(X)$, with $Y \in \boldsymbol{Y}$, s.t. $X \stackrel{s}{\leq 2} \boldsymbol{Y}$.</i>
2010.14265_F00133	6	■ 0.491	$Y \stackrel{s}{\rightarrow} \{X, Z\}$	$Y \stackrel{s}{\rightarrow} \{X, Z\}$	$Y \stackrel{s}{\rightarrow} \{X, Z\}$, two parents, while $X \stackrel{s}{\rightarrow} \{Y, Z\}$ —i.e. a child and

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2010.14265_F00134	6	■ 0.491	$X \frac{s}{2_2} \{Y, Z\}$	$X \frac{s}{2_2} \{Y, Z\}$	$Y \xrightarrow{s} \{X, Z\}$, two parents, while $X \xrightarrow{s} \{Y, Z\}$ —i.e. a child and
2010.14265_F00135	6	■ 1.000	$X \rightarrow Z \rightarrow W \rightarrow Y$	$X \rightarrow Z \rightarrow W \rightarrow Y$	$X \rightarrow Y$ and $X \rightarrow Z \rightarrow W \rightarrow Y$ that cancel such that
2010.14265_F00136	6	■ 1.000	$X \perp_P Y, X \perp_P W \mid Z, X$	$X \perp_P Y, X \perp_P W \mid Z, X$	$X \perp_P Y$. Since $X \perp_P Y, X \perp_P W \mid Z, X$ is not 1-associated
2010.14265_F00137	6	■ 0.968	W	W	to Y or W and hence could be strictly 2-associated to the
2010.14265_F00138	6	■ 0.749	$\{W, Y\}$	$\{W, Y\}$	set $\{W, Y\}$. Since we know that a 2-association contains a
2010.14265_F00139	6	■ 0.467	$Y \frac{s}{-2} \{X, Z\}$	$Y \frac{s}{-2} \{X, Z\}$	Figure 4: In both distributions $Y \xrightarrow{s} \{X, Z\}$ and $Y \rightarrow_1 W$. In
2010.14265_F00140	6	■ 0.467	$Y \rightarrow_1 W$	$Y \rightarrow_1 W$	Figure 4: In both distributions $Y \xrightarrow{s} \{X, Z\}$ and $Y \rightarrow_1 W$. In
2010.14265_F00141	6	■ 0.969	$\{X, Z\}$	$\{X, Z\}$	$\{X, Z\}$ and W , whereas in (b) Y is a non-collider.
2010.14265_F00142	6	■ 1.000	$P(W=1)=p$	$P(W = 1) = p$	variable W with $P(W = 1) = p$, where $0 < p < 1$ and an
2010.14265_F00143	6	■ 1.000	$0 < p < 1$	$0 < p < 1$	variable W with $P(W = 1) = p$, where $0 < p < 1$ and an
2010.14265_F00144	6	■ 1.000	$P(E=1)=q$	$P(E = 1) = q$	unobserved binary noise variable E with $P(E = 1) = q$,
2010.14265_F00145	6	■ 1.000	$0 < q < \frac{1}{2}$	$0 < q < \frac{1}{2}$	where $0 < q < \frac{1}{2}$. Now we generate Y as
2010.14265_F00146	6	■ 0.998	E, W, X	E, W, X	where E, W, X and Z are drawn independently. The require-
2010.14265_F00147	6	■ 0.998	$(q \neq 0)$	$(q \neq 0)$	deterministic ($q \neq 0$) and evident without observing E
2010.14265_F00148	6	■ 1.000	$\left(q \neq \frac{1}{2}\right)$	$\left(q \neq \frac{1}{2}\right)$	$(q \neq \frac{1}{2})$. The corresponding causal graph is given in Fig-
2010.14265_F00149	6	■ 1.000	$\{X, Z\} \perp_P W$	$\{X, Z\} \perp_P W$	triple, but only Y is dependent on W , whereas $\{X, Z\} \perp_P W$
2010.14265_F00150	6	■ 1.000	$X \perp_P W \mid Y$	$X \perp_P W \mid Y$	and due to the xor, $X \perp_P W \mid Y$ as well as $Z \perp_P W \mid Y$. Thus,
2010.14265_F00151	6	■ 1.000	$Z \perp_P W \mid Y$	$Z \perp_P W \mid Y$	and due to the xor, $X \perp_P W \mid Y$ as well as $Z \perp_P W \mid Y$. Thus,
2010.14265_F00152	6	■ 0.412	$X \perp_P W \mid \{Y, Z\}$	$X \perp_P W \mid \{Y, Z\}$	ever, we do find that $X \not\perp_P W \mid \{Y, Z\}$ and $Z \not\perp_P W \mid \{Y, X\}$.
2010.14265_F00153	6	■ 0.412	$Z, \perp_P W \mid \{Y, X\}$	$Z, \perp_P W \mid \{Y, X\}$	ever, we do find that $X \not\perp_P W \mid \{Y, Z\}$ and $Z \not\perp_P W \mid \{Y, X\}$.
2010.14265_F00154	6	■ 1.000	$W \rightarrow Y$	$W \rightarrow Y$	identify that Y is the collider in the triple and that $W \rightarrow Y$.
2010.14265_F00155	6	■ 0.475	$X \nVdash_P W \mid Z$	$X \nVdash_P W \mid Z$	of the edges as for example we would find that $X \nVdash_P W \mid Z$,
2010.14265_F00156	6	■ 1.000	$\boldsymbol{Y} \rightarrow X$	$\boldsymbol{Y} \rightarrow X$	write $\boldsymbol{Y} \rightarrow X$, if for each element $Y \in \boldsymbol{Y}$ it holds that $Y \rightarrow X$
2010.14265_F00157	6	■ 1.000	$Y \rightarrow X$	$Y \rightarrow X$	write $\boldsymbol{Y} \rightarrow X$, if for each element $Y \in \boldsymbol{Y}$ it holds that $Y \rightarrow X$
2010.14265_F00158	6	■ 1.000	$X \rightarrow \boldsymbol{Y}$	$X \rightarrow \boldsymbol{Y}$	and vice versa write $X \rightarrow \boldsymbol{Y}$ if X is a parent of each node
2010.14265_F00159	6	■ 1.000	$\forall Y \in \boldsymbol{Y} : X \rightarrow Y$	$\forall Y \in \boldsymbol{Y} : X \rightarrow Y$	$Y \in \boldsymbol{Y}$, that is, $\forall Y \in \boldsymbol{Y} : X \rightarrow Y$.
2010.14265_F00160	6	■ 0.902	$M := (G, V, P)$	$M := (G, V, P)$	Definition 8 (Orientation Rule) Let $M := (G, V, P)$ and
2010.14265_F00161	6	■ 0.999	$\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	$\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	we are given two disjoint sets $\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$ and $Y \in \boldsymbol{V}$, where
2010.14265_F00162	6	■ 0.999	$Y \in \boldsymbol{V}$	$Y \in \boldsymbol{V}$	we are given two disjoint sets $\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$ and $Y \in \boldsymbol{V}$, where
2010.14265_F00163	7	■ 0.998	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{X}$	$Y \xrightarrow{s}_{\leq 2} \boldsymbol{X}$	$Y \xrightarrow{s}_{\leq 2} \boldsymbol{X}$ and $Y \xrightarrow{s}_{\leq 2} \boldsymbol{Z}$, and no node $X \in \boldsymbol{X}$ is adjacent to
2010.14265_F00164	7	■ 0.998	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{Z}$	$Y \xrightarrow{s}_{\leq 2} \boldsymbol{Z}$	$Y \xrightarrow{s}_{\leq 2} \boldsymbol{X}$ and $Y \xrightarrow{s}_{\leq 2} \boldsymbol{Z}$, and no node $X \in \boldsymbol{X}$ is adjacent to
2010.14265_F00165	7	■ 0.974	$Z \in \boldsymbol{Z}$	$Z \in \boldsymbol{Z}$	some node $Z \in \boldsymbol{Z}$.
2010.14265_F00166	7	■ 0.954	$Z \in \boldsymbol{Z}$	$Z \in \boldsymbol{Z}$	i) If for each pair $X \in \boldsymbol{X}$ and $Z \in \boldsymbol{Z}$ it holds that X
2010.14265_F00167	7	■ 1.000	$Y \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	$Y \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	contains $Y \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$, then $\boldsymbol{X} \rightarrow Y \leftarrow \boldsymbol{Z}$,
2010.14265_F00168	7	■ 1.000	$\boldsymbol{X} \rightarrow Y \leftarrow \boldsymbol{Z}$	$\boldsymbol{X} \rightarrow Y \leftarrow \boldsymbol{Z}$	contains $Y \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$, then $\boldsymbol{X} \rightarrow Y \leftarrow \boldsymbol{Z}$,
2010.14265_F00169	7	■ 0.998	$\boldsymbol{V} \setminus \{X, Z\}$	$\boldsymbol{V} \setminus \{X, Z\}$	of $\boldsymbol{V} \setminus \{X, Z\}$ that contains $(\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$ but does
2010.14265_F00170	7	■ 0.998	$(\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	$(\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	of $\boldsymbol{V} \setminus \{X, Z\}$ that contains $(\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$ but does
2010.14265_F00171	7	■ 1.000	Y, Y	Y, Y	not contain Y , Y is a non-collider on at least one path
2010.14265_F00172	7	■ 1.000	$X \leftarrow Y \rightarrow Z$	$X \leftarrow Y \rightarrow Z$	$X \leftarrow Y \rightarrow Z$. Let us consider Example 2 again, where
2010.14265_F00173	7	■ 0.698	$Y \frac{s}{2_2} \{X, Z\}, Y \rightarrow_1 W$	$Y \frac{s}{2_2} \{X, Z\}, Y \rightarrow_1 W$	W . First, we find that $Y \xrightarrow{s} \{X, Z\}$, $Y \rightarrow_1 W$ and W is not
2010.14265_F00174	7	■ 1.000	$\{Y, Z\}$	$\{Y, Z\}$	that includes $\{Y, Z\}$ and W is dependent on Z given any set
2010.14265_F00175	7	■ 1.000	$\{Y, X\}$	$\{Y, X\}$	that includes $\{Y, X\}$. Hence, we can infer the correct DAG
2010.14265_F00176	7	■ 1.000	$\{X, Z\} \rightarrow Y \leftarrow W$	$\{X, Z\} \rightarrow Y \leftarrow W$	structure $\{X, Z\} \rightarrow Y \leftarrow W$.
2010.14265_F00177	7	■ 0.976	$X \not\perp_P Z$	$X \not\perp_P Z$	ture such as $X \rightarrow Y \rightarrow Z$, will we always find that $X \not\perp_P Z$?
2010.14265_F00178	7	■ 0.393	$X \nVdash_P Y$	$X \nVdash_P Y$	the true graph is $X \rightarrow Y \rightarrow Z$ and it holds that $X \not\perp_P Y$
2010.14265_F00179	7	■ 0.376	$Y, \angle_P Z$	$Y, \angle_P Z$	and $Y \not\perp_P Z$, which is called failure of transitivity. More
2010.14265_F00180	7	■ 0.913	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{X}$	$Y \xrightarrow{s}_{\leq 2} \boldsymbol{X}$	$\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$ and $Y \in \boldsymbol{V}$, where $Y \xrightarrow{s}_{\leq 2} \boldsymbol{X}$ and $Y \xrightarrow{s}_{\leq 2} \boldsymbol{Z}$, and no

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2010.14265_F00181	7	■ 0.913	$Y \underline{-}_{\leq 2}^{\{s\}} \boldsymbol{Z}$	$Y \overset{s}{\leftarrow}_{\leq 2} Z$	$X, Z \subseteq V$ and $Y \in V$, where $Y \overset{s}{\leftarrow}_{\leq 2} X$ and $Y \overset{s}{\leftarrow}_{\leq 2} Z$, and no
2010.14265_F00182	7	■ 0.985	$\boldsymbol{Z} \in \boldsymbol{Z}$	$Z \in \mathbf{Z}$	node $X \in \mathbf{X}$ is adjacent to some node $Z \in \mathbf{Z}$.
2010.14265_F00183	7	■ 0.997	$Y \stackrel{s}{\leftarrow}_{\leq 2} \boldsymbol{X} = \left\{ X_1, X_2 \right\}, Y \stackrel{s}{\leftarrow}_{\leq 2} \boldsymbol{Z} =$	$Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{X} = \{X_1, X_2\}, Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{Z} =$	Figure 5: In both figures, $Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{X} = \{X_1, X_2\}, Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{Z} =$
2010.14265_F00184	7	■ 1.000	$\left\{ Z_1, Z_2 \right\}$	$\{Z_1, Z_2\}$	$\{Z_1, Z_2\}$ (related nodes and edges are marked in black) and
2010.14265_F00185	7	■ 0.730	$X_2 \overset{s}{\leftarrow}_{\leq 2} \{U, Z_2\}$	$X_2 \overset{s}{\leftarrow}_{\leq 2} \{U, Z_2\}$	$X_2 \overset{s}{\leftarrow}_{\leq 2} \{U, Z_2\}$. Given only this information, we cannot tell if
2010.14265_F00186	7	■ 1.000	$\left\langle X_2, Y, Z_2 \right\rangle$	$\langle X_2, Y, Z_2 \rangle$	the path $\langle X_2, Y, Z_2 \rangle$ is unshielded (a) or shielded (b). While
2010.14265_F00187	7	■ 1.000	X_1	X_1	path from X_1 over U to Z_2 and the adjacency between X_2
2010.14265_F00188	7	■ 1.000	Z_2	Z_2	path from X_1 over U to Z_2 and the adjacency between X_2
2010.14265_F00189	7	■ 1.000	X_2	X_2	path from X_1 over U to Z_2 and the adjacency between X_2
2010.14265_F00190	7	■ 0.983	$X, Z \in \boldsymbol{X} \times \boldsymbol{Z}$	$X, Z \in \mathbf{X} \times \mathbf{Z}$	and Z_2 , each pair $X, Z \in \mathbf{X} \times \mathbf{Z}$ is now d -connected given
2010.14265_F00191	7	■ 1.000	$\{Y\} \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	$\{Y\} \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$	$\{Y\} \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$. Thus, the condition for rule i) could
2010.14265_F00192	7	■ 0.773	$Z \in Z, X$	$Z \in Z, X$	and $Z \in \mathbf{Z}$, X is dependent on Z given any subset of
2010.14265_F00193	7	■ 0.997	$Z \in \boldsymbol{Z}, X$	$Z \in \mathbf{Z}, X$	ii) otherwise, for each pair $X \in \mathbf{X}$ and $Z \in \mathbf{Z}$, X is depen-
2010.14265_F00194	7	■ 1.000	$(X, Z) \in \boldsymbol{X} \times \boldsymbol{Z}$	$(X, Z) \in \mathbf{X} \times \mathbf{Z}$	with $(X, Z) \in \mathbf{X} \times \mathbf{Z}$ are unshielded, we can detect if any of
2010.14265_F00195	8	■ 1.000	$\{Z, U\}$	$\{Z, U\}$	2-association between X and $\{Z, U\}$ there is no guarantee
2010.14265_F00196	8	■ 0.719	$X \rightarrow U \leftarrow Z$	$X \rightarrow U \leftarrow Z$	$X \rightarrow U \leftarrow Z$ and after that also apply our rule for $\{X, Y, Z\}$.
2010.14265_F00197	8	■ 1.000	$M := (G, \boldsymbol{V}, P)$	$M := (G, \mathbf{V}, P)$	Corollary 1 Given $M := (G, \mathbf{V}, P)$ with $Y \in \mathbf{V}$ and $X, Z \subseteq$
2010.14265_F00198	8	■ 1.000	$\boldsymbol{X}, \boldsymbol{Z} \subseteq$	$\mathbf{X}, \mathbf{Z} \subseteq$	Corollary 1 Given $M := (G, \mathbf{V}, P)$ with $Y \in \mathbf{V}$ and $X, Z \subseteq$
2010.14265_F00199	8	■ 0.280	$\boldsymbol{X} \cap \boldsymbol{Z} = \emptyset, Y \overset{s}{\leftarrow}_{\leq 2} \boldsymbol{X}, Y \overset{s}{\leftarrow}_{\leq 2} \boldsymbol{Z}$	$\mathbf{X} \cap \mathbf{Z} = \emptyset, Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{X}, Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{Z}$	$\mathbf{V}$, where $\mathbf{X} \cap \mathbf{Z} = \emptyset, Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{X}, Y \overset{s}{\leftarrow}_{\leq 2} \mathbf{Z}$ and no pair of nodes
2010.14265_F00200	8	■ 0.923	$(X, Z) \in X \times Z$	$(X, Z) \in X \times Z$	$(X, Z) \in \mathbf{X} \times \mathbf{Z}$ is adjacent. Assuming that CMC holds, we
2010.14265_F00201	8	■ 0.999	$\{\boldsymbol{X}, Y, \boldsymbol{Z}\}$	$\{\mathbf{X}, Y, \mathbf{Z}\}$	fails on the triple $\{X, Y, Z\}$.
2010.14265_F00202	8	■ 1.000	T	T	a target node T , and a shrink phase, in which superfluous
2010.14265_F00203	8	■ 1.000	$T \rightarrow C \leftarrow S$	$T \rightarrow C \leftarrow S$	ness. In particular, consider the graph $T \rightarrow C \leftarrow S$, in which
2010.14265_F00204	8	■ 0.916	$T \overset{s}{\leftarrow}_{\leq 2} Y$	$T \overset{s}{\leftarrow}_{\leq 2} Y$	$T \overset{s}{\leftarrow}_{\leq 2} Y$ and $C \overset{s}{\leftarrow}_{\leq 2} Y$. Assuming only 2-adjacency faithfulness
2010.14265_F00205	8	■ 0.916	$C \overset{s}{\leftarrow}_{\leq 2} Y$	$C \overset{s}{\leftarrow}_{\leq 2} Y$	$T \overset{s}{\leftarrow}_{\leq 2} Y$ and $C \overset{s}{\leftarrow}_{\leq 2} Y$. Assuming only 2-adjacency faithfulness
2010.14265_F00206	8	■ 1.000	$T \not\perp S \mid C$	$T \not\perp S \mid C$	will not guarantee that $T \not\perp S \mid C$, additionally assuming
2010.14265_F00207	8	■ 0.998	$\boldsymbol{X} = \{X, T\}$	$\mathbf{X} = \{X, T\}$	$\mathbf{X} = \{X, T\}$, where T is the target node. Then all nodes in
2010.14265_F00208	8	■ 0.980	$\mathrm{PC}(T)$	$\mathrm{PC}(T)$	holds and even become part of $\mathrm{PC}(T)$ if all paths $\langle T, Y, Z \rangle$
2010.14265_F00209	8	■ 0.980	$\langle T, Y, Z \rangle$	$\langle T, Y, Z \rangle$	holds and even become part of $\mathrm{PC}(T)$ if all paths $\langle T, Y, Z \rangle$
2010.14265_F00210	8	■ 1.000	$T \rightarrow Y \leftarrow Z$	$T \rightarrow Y \leftarrow Z$	consists of the paths $T \rightarrow Y \leftarrow Z$ and $Y \leftarrow X \rightarrow Z$. Since X
2010.14265_F00211	8	■ 1.000	$Y \leftarrow X \rightarrow Z$	$Y \leftarrow X \rightarrow Z$	consists of the paths $T \rightarrow Y \leftarrow Z$ and $Y \leftarrow X \rightarrow Z$. Since X
2010.14265_F00212	8	■ 0.514	$T \overset{s}{\leftarrow}_{\leq 2} \{X, Z\}$	$T \overset{s}{\leftarrow}_{\leq 2} \{X, Z\}$	For example, assume that $T \overset{s}{\leftarrow}_{\leq 2} \{X, Z\}$ and both nodes were
2010.14265_F00213	8	■ 0.380	$T \not\perp_P X \mid S \setminus X$	$T \not\perp_P X \mid S \setminus X$	phase, we find that $T \not\perp_P X \mid S \setminus X$, since $Z \in S$. Hence, X
2010.14265_F00214	8	■ 0.380	$Z \in \boldsymbol{S}$	$Z \in \mathbf{S}$	phase, we find that $T \not\perp_P X \mid S \setminus X$, since $Z \in S$. Hence, X
2010.14265_F00215	8	■ 0.986	$\boldsymbol{S}$	$\mathbf{S}$	remains in $\mathbf{S}$, as well as Z .
2010.14265_F00216	8	■ 0.936	$M = (G, V, P)$	$M = (G, V, P)$	Theorem 3 Given $M = (G, V, P)$. Assuming that 2-
2010.14265_F00217	8	■ 0.981	$MB(T)$	$MB(T)$	rithm 1 correctly identifies $MB(T)$ for $T \in V$.
2010.14265_F00218	8	■ 0.981	$T \in \boldsymbol{V}$	$T \in \mathbf{V}$	rithm 1 correctly identifies $MB(T)$ for $T \in V$.
2010.14265_F00219	9	■ 0.924	$\mathrm{MB}(T)$	$\mathbf{MB}(T)$	in which case or
2010.14265_F00220	9	■ 0.917	$\boldsymbol{V}' \leftarrow \boldsymbol{V} \setminus \{T\}$	$\mathbf{V}' \leftarrow \mathbf{V} \setminus \{T\}$	$\mathbf{V}' \leftarrow \mathbf{V} \setminus \{T\}$;
2010.14265_F00221	9	■ 0.995	$\boldsymbol{S} \leftarrow \emptyset$	$\mathbf{S} \leftarrow \emptyset$	$\mathbf{S} \leftarrow \emptyset$;
2010.14265_F00222	9	■ 0.836	$\left(\exists X \in \boldsymbol{V}' : T \not\perp_P X \mid \boldsymbol{S} \right) \vee$	$(\exists X \in \mathbf{V}' : T \not\perp_P X \mid \mathbf{S}) \vee$	while $(\exists X \in \mathbf{V}' : T \not\perp_P X \mid \mathbf{S}) \vee$
2010.14265_F00223	9	■ 0.716	$\left(\exists X, Z \in \boldsymbol{V}' : T \not\perp_P X \mid \boldsymbol{S} \cup \{Z\} \right) \text{ do}$	$(\exists X, Z \in \mathbf{V}' : T \not\perp_P X \mid \mathbf{S} \cup \{Z\})$	$(\exists X, Z \in \mathbf{V}' : T \not\perp_P X \mid \mathbf{S} \cup \{Z\})$ do
2010.14265_F00224	9	■ 1.000	$\boldsymbol{S} \leftarrow \boldsymbol{S} \cup \{X\}$	$\mathbf{S} \leftarrow \mathbf{S} \cup \{X\}$	$\mathbf{S} \leftarrow \mathbf{S} \cup \{X\}$;

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2010.14265_F00225	9	0.788	$\exists X \in \mathcal{S} : T \perp_P X \mid \mathcal{S} \backslash X$	$\exists X \in \mathcal{S} : T \perp_P X \mid \mathcal{S} \backslash X$	while $\exists X \in \mathcal{S} : T \perp_P X \mid \mathcal{S} \backslash X$ do
2010.14265_F00226	9	1.000	$\boldsymbol{\mathcal{S}} \leftarrow \boldsymbol{\mathcal{S}} \backslash X ;$	$\mathcal{S} \leftarrow \mathcal{S} \backslash X ;$	$\mathcal{S} \leftarrow \mathcal{S} \backslash X ;$
2010.14265_F00227	9	1.000	$X \rightarrowtail Y \leftarrowtail Z, Y \rightarrowtail X \leftarrowtail Z$	$X \rightarrow Y \leftarrow Z, Y \rightarrow X \leftarrow Z$	$X \rightarrow Y \leftarrow Z, Y \rightarrow X \leftarrow Z$ and $X \rightarrow Z \leftarrow Y$, also a
2010.14265_F00228	9	1.000	$X \rightarrowtail Z$	$X \rightarrow Z$	fully connected graph, e.g. $X \rightarrow Y \leftarrow Z$ and $X \rightarrow Z$, is
2010.14265_F00229	11	0.464	$W \perp_P Y$	$W \perp_P Y$	mal unfaithful triple and $W \not\perp_P Y$ as follows. We generate
2010.14265_F00230	11	1.000	X, Z, W	X, Z, W	X, Z, W and E independently, with X and Z as fair coins,
2010.14265_F00231	11	1.000	$q > 0$	$q > 0$	$0 < q < \frac{1}{2}$. With $q > 0$, we ensure that the function is
2010.14265_F00232	11	1.000	$P(Y=1)=q+\frac{p}{2}-pq$	$P(Y = 1) = q + \frac{p}{2} - pq$	We will obtain that $P(Y = 1) = q + \frac{p}{2} - pq$. Further, we
2010.14265_F00233	11	1.000	$P(X=1, Y=1)=\frac{1}{2} P(Y=1)=P(X=$	$P(X = 1, Y = 1) = \frac{1}{2} P(Y = 1) = P(X =$	can calculate that $P(X = 1, Y = 1) = \frac{1}{2} P(Y = 1) = P(X =$
2010.14265_F00234	11	0.958	$1) \cdot P(Y=1)$	$1) \cdot P(Y = 1)$	$1) \cdot P(Y = 1)$. Also, $P(X = 1, Y = 0) = P(X = 1) \cdot P(Y = 0)$,
2010.14265_F00235	11	0.958	$P(X=1, Y=0)=P(X=1) \cdot P(Y=0)$	$P(X = 1, Y = 0) = P(X = 1) \cdot P(Y = 0)$	$1) \cdot P(Y = 1)$. Also, $P(X = 1, Y = 0) = P(X = 1) \cdot P(Y = 0)$,
2010.14265_F00236	11	1.000	$P(X=0, Z=1, Y=1)=$	$P(X = 0, Z = 1, Y = 1) =$	all three variables, we get that $P(X = 0, Z = 1, Y = 1) =$
2010.14265_F00237	11	1.000	$\frac{p+q-2pq}{4}$	$\frac{p+q-2pq}{4}$	$\frac{p+q-2pq}{4}$ and $P(X = 0, Z = 1) \cdot P(Y = 1) = \frac{1}{4} P(Y = 1)$.
2010.14265_F00238	11	1.000	$P(X=0, Z=1) \cdot P(Y=1)=\frac{1}{4} P(Y=1)$	$P(X = 0, Z = 1) \cdot P(Y = 1) = \frac{1}{4} P(Y = 1)$	$\frac{p+q-2pq}{4}$ and $P(X = 0, Z = 1) \cdot P(Y = 1) = \frac{1}{4} P(Y = 1)$.
2010.14265_F00239	11	1.000	$p=0$	$p = 0$	The only solutions are $p = 0$ or $q = \frac{1}{2}$, which we excluded.
2010.14265_F00240	11	1.000	$q=\frac{1}{2}$	$q = \frac{1}{2}$	The only solutions are $p = 0$ or $q = \frac{1}{2}$, which we excluded.
2010.14265_F00241	11	0.462	$Y \not\perp_P X \mid Z$	$Y \not\perp_P X \mid Z$	Hence, $Y \not\perp_P \{X, Z\}$ and by weak union also $Y \not\perp_P X \mid Z$,
2010.14265_F00242	11	0.870	$Y \not\perp_P Z \mid X$	$Y \not\perp_P Z \mid X$	as well as $Y \not\perp_P Z \mid X$. Since we know by assumption
2010.14265_F00243	11	0.288	$W \mathcal{H}_P Y$	$W \mathcal{H}_P Y$	$W \not\perp_P Y$, since $P(W = 1, Y = 1) = \frac{p}{2}$, which is only equal
2010.14265_F00244	11	0.288	$P(W=1, Y=1)=\frac{p}{2}$	$P(W = 1, Y = 1) = \frac{p}{2}$	$W \not\perp_P Y$, since $P(W = 1, Y = 1) = \frac{p}{2}$, which is only equal
2010.14265_F00245	11	1.000	$P(W=1) \cdot P(Y=1)$	$P(W = 1) \cdot P(Y = 1)$	to $P(W = 1) \cdot P(Y = 1)$, if $p = 0$, $p = 1$ or $q = \frac{1}{2}$, which we
2010.14265_F00246	11	1.000	$p=0, p=1$	$p = 0, p = 1$	to $P(W = 1) \cdot P(Y = 1)$, if $p = 0$, $p = 1$ or $q = \frac{1}{2}$, which we
2010.14265_F00247	11	0.396	$X \measuredangle_P W \mid \{Y, Z\}$	$X \angle_P W \mid \{Y, Z\}$	$X \not\perp_P W \mid \{Y, Z\}$ and that $Z \not\perp_P W \mid \{X, Y\}$. We can write
2010.14265_F00248	11	0.396	$Z \not\perp_P W \mid \{X, Y\}$	$Z \not\perp_P W \mid \{X, Y\}$	$X \not\perp_P W \mid \{Y, Z\}$ and that $Z \not\perp_P W \mid \{X, Y\}$. We can write
2010.14265_F00249	11	1.000	$P(X \mid Y, Z) \cdot P(W \mid Y, Z)$	$P(X \mid Y, Z) \cdot P(W \mid Y, Z)$	ent from $P(X \mid Y, Z) \cdot P(W \mid Y, Z)$. Consider the case where
2010.14265_F00250	11	1.000	$P(Y=1, Z=1)=P(Y=1) / 2$	$P(Y = 1, Z = 1) = P(Y = 1)/2$	Since we know that $P(Y = 1, Z = 1) = P(Y = 1)/2$, we thus
2010.14265_F00251	11	0.998	$q=0$	$q = 0$	This equation can only be true if p or $q = 0$, i.e. the system
2010.14265_F00252	11	1.000	$p=1$	$p = 1$	is either independent of W or deterministic, $p = 1$ or $q = \frac{1}{2}$,
2010.14265_F00253	11	0.453	$X \perp_P W$	$X \perp_P W$	which we all excluded by assumption. Hence, $X \not\perp_P W \mid$
2010.14265_F00254	11	1.000	$G^{\ast}$	$G^{\ast}$	discussed in the related work section. A graph $G^{\ast}$ satisfies
2010.14265_F00255	11	0.999	G_{π}	G_{π}	To explain the SP algorithm, we need to define a DAG G_{π} ,
2010.14265_F00256	11	0.999	π	π	w.r.t. a permutation π . A DAG G_{π} consists of vertices V and
2010.14265_F00257	11	1.000	E_{π}	E_{π}	directed edges E_{π} , where an edge from the j -th node $\pi(j)$
2010.14265_F00258	11	1.000	j	j	directed edges E_{π} , where an edge from the j -th node $\pi(j)$
2010.14265_F00259	11	1.000	$\pi(j)$	$\pi(j)$	directed edges E_{π} , where an edge from the j -th node $\pi(j)$
2010.14265_F00260	11	1.000	$\pi(k)$	$\pi(k)$	according to permutation π to node $\pi(k)$ is in E_{π} if and only
2010.14265_F00261	11	1.000	$j < k$	$j < k$	if $j < k$ and
2010.14265_F00262	11	0.999	$X_{\pi(j)}$	$X_{\pi(j)}$	where $X_{\pi(j)}$ refers to the j -th random variable according to
2010.14265_F00263	11	1.000	$\pi^{\ast}$	$\pi^{\ast}$	lects that permutation $\pi^{\ast}$ for which $G_{\pi^{\ast}}$ contains the fewest
2010.14265_F00264	11	1.000	$G_{\pi^{\ast}}$	$G_{\pi^{\ast}}$	lects that permutation $\pi^{\ast}$ for which $G_{\pi^{\ast}}$ contains the fewest
2010.14265_F00265	11	0.874	$\pi(4)=Y$	$\pi(4) = Y$	$\pi(4) = Y$. W.l.o.g. let $\pi(1) = X, \pi(2) = Z$ and $\pi(3) = W$,
2010.14265_F00266	11	0.874	$\pi(1)=X, \pi(2)=Z$	$\pi(1) = X, \pi(2) = Z$	$\pi(4) = Y$. W.l.o.g. let $\pi(1) = X, \pi(2) = Z$ and $\pi(3) = W$,
2010.14265_F00267	11	0.874	$\pi(3)=W$	$\pi(3) = W$	$\pi(4) = Y$. W.l.o.g. let $\pi(1) = X, \pi(2) = Z$ and $\pi(3) = W$,
2010.14265_F00268	12	1.000	$\pi^{\prime}$	$\pi^{\prime}$	If we would pick a permutation $\pi^{\prime}$ in which we flip for
2010.14265_F00269	12	1.000	$\pi^{\prime}(3)=Y$	$\pi^{\prime}(3) = Y$	to the highest number in the permutation, i.e. $\pi^{\prime}(3) = Y$ and
2010.14265_F00270	12	0.999	$\pi^{\prime}(4)=W$	$\pi^{\prime}(4) = W$	$\pi^{\prime}(4) = W$, we will find more edges and thus not a minimal

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2010.14265_F00271	12	■ 1.000	$\pi^{\prime}(1)=Z, \pi^{\prime}(2)=$	$\pi'(1) = Z, \pi'(2) =$	put X as the last node and get the order $\pi'(1) = Z, \pi'(2) =$
2010.14265_F00272	12	■ 1.000	$W, \pi^{\prime}(3)=Y$	$W, \pi'(3) = Y$	$W, \pi'(3) = Y$ and $\pi'(4) = X$, for which
2010.14265_F00273	12	■ 1.000	$\pi^{\prime}(4)=X$	$\pi'(4) = X$	$W, \pi'(3) = Y$ and $\pi'(4) = X$, for which
2010.14265_F00274	12	■ 1.000	$\pi^{\prime}(1)=Y$	$\pi'(1) = Y$	for π . Also, if $\pi'(1) = Y$, we can use it in the conditional
2010.14265_F00275	12	■ 1.000	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	with $\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$. The (semi-)graphoid axioms are the
2010.14265_F00276	12	■ 0.979	$\left(\perp\right)$	$(\perp)$	following rules ($\perp$ denotes $\perp_P$ and $\perp_G$)
2010.14265_F00277	12	■ 0.979	$\perp_P$	$\perp_P$	following rules ($\perp$ denotes $\perp_P$ and $\perp_G$)
2010.14265_F00278	12	■ 0.979	$\left(\perp_G\right)$	$\perp_G$	following rules ($\perp$ denotes $\perp_P$ and $\perp_G$)
2010.14265_F00279	12	■ 1.000	$\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z} \Rightarrow \boldsymbol{Y} \perp \boldsymbol{X} \boldsymbol{Z}$	$\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z} \Rightarrow \boldsymbol{Y} \perp \boldsymbol{X} \boldsymbol{Z}$	1. Symmetry: $\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z} \Rightarrow \boldsymbol{Y} \perp \boldsymbol{X} \boldsymbol{Z}$.
2010.14265_F00280	12	■ 1.000	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}$	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}$	2. Decomposition: $\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}$.
2010.14265_F00281	12	■ 1.000	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}$	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}$	3. Weak Union: $\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}$.
2010.14265_F00282	12	■ 1.000	$(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Z}) \Rightarrow$	$(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Z}) \Rightarrow$	4. Contraction: $(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Z}) \Rightarrow$
2010.14265_F00283	12	■ 0.570	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z}$	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z}$	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z}$.
2010.14265_F00284	12	■ 1.000	$\perp_G$	$\perp_G$	include two additional rules (only for $\perp_G$).
2010.14265_F00285	12	■ 1.000	$(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Y} \cup \boldsymbol{Z}) \Rightarrow$	$(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Y} \cup \boldsymbol{Z}) \Rightarrow$	5. Intersection: $(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Y} \cup \boldsymbol{Z}) \Rightarrow$
2010.14265_F00286	12	■ 1.000	$(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Z}) \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \cup$	$(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Z}) \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \cup$	6. Composition: $(\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \boldsymbol{Z}) \Rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \cup$
2010.14265_F00287	12	■ 0.999	$\boldsymbol{W} \boldsymbol{Z}$	$\boldsymbol{W} \boldsymbol{Z}$	$\boldsymbol{W} \boldsymbol{Z}$.
2010.14265_F00288	12	■ 1.000	$Y \perp_P X$	$Y \perp_P X$	by the xor, we find that $Y \perp_P X$ and $Y \perp_P Z$ but we cannot
2010.14265_F00289	12	■ 1.000	$Y \perp_P Z$	$Y \perp_P Z$	by the xor, we find that $Y \perp_P X$ and $Y \perp_P Z$ but we cannot
2010.14265_F00290	12	■ 1.000	$Y \perp_P \{X, Z\}$	$Y \perp_P \{X, Z\}$	conclude that $Y \perp_P \{X, Z\}$. If, however, in a graph Y is d -
2010.14265_F00291	12	■ 0.942	$X \not\perp_P \{Y, Z\}, Y \not\perp_P \{X, Z\}$	$X \not\perp_P \{Y, Z\}, Y \not\perp_P \{X, Z\}$	an unfaithful triple in P , then $X \not\perp_P \{Y, Z\}, Y \not\perp_P \{X, Z\}$ and
2010.14265_F00292	12	■ 0.970	$X \not\perp_P \{Y, Z\}$	$X \not\perp_P \{Y, Z\}$	Proof: Assume that w.l.o.g. $X \not\perp_P \{Y, Z\}$ is violated. By
2010.14265_F00293	12	■ 1.000	$Y \perp_P X \mid Z$	$Y \perp_P X \mid Z$	$Y \perp_P X \mid Z$, using symmetry. We know that $Y \perp_P Z$. By
2010.14265_F00294	12	■ 0.999	$X \perp_P \{Y, Z\}$	$X \perp_P \{Y, Z\}$	e.g. $X \not\perp_P \{Y, Z\}$ and $X \perp_P Z$, implies a conditional depen-
2010.14265_F00295	12	■ 0.593	$X \not\perp_P Y \mid Z$	$X \not\perp_P Y \mid Z$	dence, e.g. $X \not\perp_P Y \mid Z$, by contraction.
2010.14265_F00296	12	—	$\square$	$\square$	—
2010.14265_F00297	12	■ 1.000	$X \perp_G Y$	$X \perp_G Y$	we write $X \stackrel{s}{\perp}_{\leq 2} Y$, if X is 1- or strictly 2-associated to Y .
2010.14265_F00298	12	■ 1.000	$X \perp_G Z$	$X \perp_G Z$	we write $X \stackrel{s}{\perp}_{\leq 2} Y$, if X is 1- or strictly 2-associated to Y .
2010.14265_F00299	12	■ 1.000	$X \perp_G \{Y, Z\}$	$X \perp_G \{Y, Z\}$	Pairwise dependencies can occur for example in a simple
2010.14265_F00300	12	■ 0.793	$X \not\perp_P Y \mid Z \cup S, X \not\perp_P Z \mid Y \cup S$	$X \not\perp_P Y \mid Z \cup S, X \not\perp_P Z \mid Y \cup S$	mon cause structure does not induce a 2-association. On
2010.14265_F00301	12	■ 0.793	$Y \not\perp_P Z \mid X \cup S$	$Y \not\perp_P Z \mid X \cup S$	mon cause structure does not induce a 2-association. On
2010.14265_F00302	13	■ 1.000	$Z \notin \text{An}(X)$	$Z \notin \text{An}(X)$	is not an ancestor of any of the other nodes, say $Z \notin \text{An}(X)$
2010.14265_F00303	13	■ 1.000	$Z \notin \text{An}(Y)$	$Z \notin \text{An}(Y)$	and $Z \notin \text{An}(Y)$, because of acyclicity. In other words, $X \notin$
2010.14265_F00304	13	■ 1.000	$X \notin$	$X \notin$	and $Z \notin \text{An}(Y)$, because of acyclicity. In other words, $X \notin$
2010.14265_F00305	13	■ 0.999	$\text{De}(Z)$	$\text{De}(Z)$	$\text{De}(Z)$ and $Y \notin \text{De}(Z)$. The local Markov property states that
2010.14265_F00306	13	■ 0.999	$Y \notin \text{De}(Z)$	$Y \notin \text{De}(Z)$	$\text{De}(Z)$ and $Y \notin \text{De}(Z)$. The local Markov property states that
2010.14265_F00307	13	■ 0.993	$Z \perp_G \text{Nd}(Z) \mid \text{Pa}(Z)$	$Z \perp_G \text{Nd}(Z) \mid \text{Pa}(Z)$	$Z \perp_G \text{Nd}(Z) \mid \text{Pa}(Z)$ and hence in particular

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2010.14265_F00308	13	0.772	$ \operatorname{Pa}(Z) \setminus \{X, Y\} < 2$	$ \operatorname{Pa}(Z) \cap \{X, Y\} < 2$	Further, if $ \operatorname{Pa}(Z) \cap \{X, Y\} < 2$, we get a contradiction with
2010.14265_F00309	13	1.000	$\{X, Y\} \subseteq \operatorname{Pa}(Z)$	$\{X, Y\} \subseteq \operatorname{Pa}(Z)$	the assumed conditional dependences. Hence $\{X, Y\} \subseteq \operatorname{Pa}(Z)$
2010.14265_F00310	13	0.989	$\operatorname{Pa}(X) \subseteq \mathbf{V} \setminus \mathbf{Z}$	$\operatorname{Pa}(X) \subseteq \mathbf{V} \setminus \mathbf{Z}$	where $\operatorname{Pa}(X) \subseteq \mathbf{V} \setminus \mathbf{Z}$. Due to acyclicity, there has to exist a
2010.14265_F00311	13	0.884	$\mathbf{X} \cup \mathbf{Z}$	$\mathbf{X} \cup \mathbf{Z}$	node in $\mathbf{X} \cup \mathbf{Z}$, say X , that is not an ancestor of any node in
2010.14265_F00312	13	0.942	$(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\}$	$(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\}$	$(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\}$ and hence $(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \subseteq \operatorname{Nd}(X)$. By the local
2010.14265_F00313	13	0.942	$(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \subseteq \operatorname{Nd}(X)$	$(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \subseteq \operatorname{Nd}(X)$	$(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\}$ and hence $(\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \subseteq \operatorname{Nd}(X)$. By the local
2010.14265_F00314	13	0.997	$X \perp_G (\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \mid \operatorname{Pa}(X)$	$X \perp_G (\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \mid \operatorname{Pa}(X)$	Markov condition, we get that $X \perp_G (\mathbf{X} \cup \mathbf{Z}) \setminus \{X\} \mid \operatorname{Pa}(X)$.
2010.14265_F00315	13	0.914	$Z \cap \operatorname{Pa}(X) = \emptyset$	$Z \cap \operatorname{Pa}(X) = \emptyset$	for any $Z \in \mathbf{Z}$. Further, $Z \cap \operatorname{Pa}(X) = \emptyset$, as by assumption no
2010.14265_F00316	13	0.789	$\mathbf{X} \rightarrow \mathbf{Y} \leftarrow \mathbf{Z}$	$\mathbf{X} \rightarrow \mathbf{Y} \leftarrow \mathbf{Z}$	sponds to $\mathbf{X} \rightarrow \mathbf{Y} \leftarrow \mathbf{Z}$,
2010.14265_F00317	13	1.000	$\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Z\}$	$\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Z\}$	and there exists a subset $\mathbf{S} \subseteq \mathbf{V} \setminus \{X, Z\}$ s.t.
2010.14265_F00318	13	0.999	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$	for which $X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$. Since $Y \in$
2010.14265_F00319	13	0.999	$Y \in$	$Y \in$	for which $X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$. Since $Y \in$
2010.14265_F00320	13	0.922	$\operatorname{Nd}(X)$	$\operatorname{Nd}(X)$	$\operatorname{Nd}(X)$, we will also find that $X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup$
2010.14265_F00321	13	0.922	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup$	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup$	$\operatorname{Nd}(X)$, we will also find that $X \perp_G Z \mid \operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup$
2010.14265_F00322	13	1.000	$(\mathbf{Z} \setminus \{Z\}) \cup \{Y\}$	$(\mathbf{Z} \setminus \{Z\}) \cup \{Y\}$	$(\mathbf{Z} \setminus \{Z\}) \cup \{Y\}$, where $\operatorname{Pa}(X)$ does not include X or Z . Thus,
2010.14265_F00323	13	1.000	$\operatorname{De}(Y)$	$\operatorname{De}(Y)$	as previously, we show that X has to be in $\operatorname{De}(Y)$ and thus
2010.14265_F00324	13	0.658	$Y \in \operatorname{Nd}(X)$	$Y \in \operatorname{Nd}(X)$	$Y \in \operatorname{Nd}(X)$. We do this by deriving a contradiction: assume
2010.14265_F00325	13	0.976	$X \in \operatorname{Nd}(Y)$	$X \in \operatorname{Nd}(Y)$	$X \in \operatorname{Nd}(Y)$. If $\mathbf{X}$ consists only of the single node X , then X
2010.14265_F00326	13	0.992	$Y, X \in \operatorname{Pa}(Y)$	$Y, X \in \operatorname{Pa}(Y)$	has to be adjacent to Y , $X \in \operatorname{Pa}(Y)$ and hence $X \rightarrow Y$ in G .
2010.14265_F00327	13	1.000	X'	X'	$\mathbf{X}$ contains a second node, X' , we know by assumption that
2010.14265_F00328	13	0.995	$X' \in \operatorname{Nd}(X)$	$X' \in \operatorname{Nd}(X)$	$X' \in \operatorname{Nd}(X)$. We also know that the triple $\{X, X', Y\}$ has to
2010.14265_F00329	13	0.995	$\{X, X', Y\}$	$\{X, X', Y\}$	$X' \in \operatorname{Nd}(X)$. We also know that the triple $\{X, X', Y\}$ has to
2010.14265_F00330	13	1.000	$X \notin \operatorname{De}(Y)$	$X \notin \operatorname{De}(Y)$	contain a collider. X cannot be the collider, since $X \notin \operatorname{De}(Y)$
2010.14265_F00331	13	1.000	$X \notin \operatorname{An}(X')$	$X \notin \operatorname{An}(X')$	and also X' cannot be the collider since $X \notin \operatorname{An}(X')$. Hence,
2010.14265_F00332	13	1.000	$\langle X, Y, X' \rangle$	$\langle X, Y, X' \rangle$	Y has to be the collider on the path $\langle X, Y, X' \rangle$. As above,
2010.14265_F00333	13	1.000	$X \in \operatorname{An}(Z)$	$X \in \operatorname{An}(Z)$	assumption and thus, $X \in \operatorname{An}(Z)$, which is a contradiction.
2010.14265_F00334	13	0.611	$\operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$	$\operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$	$\operatorname{Pa}(X) \cup (\mathbf{X} \setminus \{X\}) \cup (\mathbf{Z} \setminus \{Z\})$. Since Y cannot be in $\operatorname{Pa}(X)$ due
2010.14265_F00335	13	0.196	$\mathbf{X} \cap \mathbf{Z} = \emptyset, Y_{-_{\leq 2}} \mathbf{X}, Y^s_{\leq 2} \mathbf{Z}$	$\mathbf{X} \cap \mathbf{Z} = \emptyset, Y_{-_{\leq 2}} \mathbf{X}, Y^s_{\leq 2} \mathbf{Z}$	The generalized GS algorithm is shown in Algorithm 1, where we only modified the grow phase to also consider
2010.14265_F00336	14	0.995	$\operatorname{MB}(T) \subseteq \mathbf{S}$	$\operatorname{MB}(T) \subseteq \mathbf{S}$	$\operatorname{MB}(T) \subseteq \mathbf{S}$ after the grow phase and second, we need to
2010.14265_F00337	14	0.862	$\operatorname{MB}(T)$	$\operatorname{MB}(T)$	ensure that all nodes in $\operatorname{MB}(T)$ stay in $\mathbf{S}$ during the shrink
2010.14265_F00338	14	0.894	$X \in \operatorname{PC}(T), T$	$X \in \operatorname{PC}(T), T$	each node $X \in \operatorname{PC}(T)$, T is either 1-associated to X , or there
2010.14265_F00339	14	0.226	$\frac{s}{2} \mathbf{X}$	$\frac{s}{2} \mathbf{X}$	exists a set $\mathbf{X}$ that includes X such that $T \xrightarrow{s} \mathbf{X}$. If T is
2010.14265_F00340	14	0.425	$T, \perp_P \mathbf{X} \mid \mathbf{S}$	$T, \perp_P \mathbf{X} \mid \mathbf{S}$	1-associated to a node X , then $T \not\perp_P \mathbf{X} \mid \mathbf{S}$, if $X \notin \mathbf{S}$, hence
2010.14265_F00341	14	0.425	$X \notin \mathbf{S}$	$X \notin \mathbf{S}$	1-associated to a node X , then $T \not\perp_P \mathbf{X} \mid \mathbf{S}$, if $X \notin \mathbf{S}$, hence
2010.14265_F00342	14	0.799	$T \not\perp_P \mathbf{X} \mid \mathbf{S} \cup \{Z\}$	$T \not\perp_P \mathbf{X} \mid \mathbf{S} \cup \{Z\}$	set $\{X, Z\}$ then $T \not\perp_P \mathbf{X} \mid \mathbf{S} \cup \{Z\}$ for all $\mathbf{S} \subseteq \mathbf{V} \setminus \{X, T, Z\}$.
2010.14265_F00343	14	0.799	$\mathbf{S} \subseteq \mathbf{V} \setminus \{X, T, Z\}$	$\mathbf{S} \subseteq \mathbf{V} \setminus \{X, T, Z\}$	set $\{X, Z\}$ then $T \not\perp_P \mathbf{X} \mid \mathbf{S} \cup \{Z\}$ for all $\mathbf{S} \subseteq \mathbf{V} \setminus \{X, T, Z\}$.
2010.14265_F00344	14	0.595	$T \perp_P \mathbf{Z} \mid \mathbf{S}$	$T \perp_P \mathbf{Z} \mid \mathbf{S}$	find that $T \not\perp_P \mathbf{Z} \mid \mathbf{S}$, if $Z \notin \mathbf{S}$, since $X \in \mathbf{S}$. Hence, all
2010.14265_F00345	14	0.595	$Z \notin \mathbf{S}$	$Z \notin \mathbf{S}$	find that $T \not\perp_P \mathbf{Z} \mid \mathbf{S}$, if $Z \notin \mathbf{S}$, since $X \in \mathbf{S}$. Hence, all
2010.14265_F00346	14	0.595	$X \in \mathbf{S}$	$X \in \mathbf{S}$	find that $T \not\perp_P \mathbf{Z} \mid \mathbf{S}$, if $Z \notin \mathbf{S}$, since $X \in \mathbf{S}$. Hence, all
2010.14265_F00347	14	0.835	(T)	(T)	with $\operatorname{PC}(T)$, hence might not have been added yet. ⁴ Since
2010.14265_F00348	14	0.947	$\mathbf{S} : 1)$	$\mathbf{S} : 1)$	particular, we need to consider two classes of spouses $\mathbf{S} : 1)$

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2010.14265_F00349	14	■ 0.563	$T \left(T \frac{s}{2_2} \{C, S\} \right)$	$T \left(T \frac{s}{2_2} \{C, S\} \right)$	to $T (T \frac{s}{2_2} \{C, S\})$. Those will be added due to the strict
2010.14265_F00350	14	■ 1.000	S	S	a conditional dependence between T and S by condition-
2010.14265_F00351	14	■ 1.000	$S_{\{1\}}$	S_1	C is strictly 2-associated to two spouses S_1 and S_2 . Due
2010.14265_F00352	14	■ 1.000	$S_{\{2\}}$	S_2	C is strictly 2-associated to two spouses S_1 and S_2 . Due
2010.14265_F00353	14	■ 0.736	$W \notin \operatorname{MB}(T)$	$W \notin \operatorname{MB}(T)$	shrink phase all $W \notin \operatorname{MB}(T)$ will be deleted from S and all
2010.14265_F00354	14	■ 0.851	$X \in \operatorname{MB}(T)$	$X \in \operatorname{MB}(T)$	$X \in \operatorname{MB}(T)$ will stay in S .
2010.14265_F00355	14	■ 0.935	$\operatorname{PC}(T)$	$\operatorname{PC}(T)$	⁴ There could be nodes that are spouses of T and in $\operatorname{PC}(T)$ at
2010.14265_F00356	14	■ 0.999	$X \in \operatorname{PC}(T)$	$X \in \operatorname{PC}(T)$	attempt to remove from S . If $X \in \operatorname{PC}(T)$, by definition of
2010.14265_F00357	14	■ 0.967	$\boldsymbol{X} \subseteq \operatorname{MB}(T)$	$\boldsymbol{X} \subseteq \operatorname{MB}(T)$	to a set $\boldsymbol{X} \subseteq \operatorname{MB}(T)$ that contains X . W.l.o.g. let $\boldsymbol{X} = \{X, Z\}$,
2010.14265_F00358	14	■ 0.967	$\boldsymbol{X} = \{X, Z\}$	$\boldsymbol{X} = \{X, Z\}$	to a set $\boldsymbol{X} \subseteq \operatorname{MB}(T)$ that contains X . W.l.o.g. let $\boldsymbol{X} = \{X, Z\}$,
2010.14265_F00359 [conf 0.097]	14	■ 0.097	$X \mid \boldsymbol{S} \backslash \{X\}$	$X \mid \boldsymbol{S} \backslash \{X\}$	then $T \not\perp_P X \mid \boldsymbol{S} \backslash \{X\}$, since S contains Z , and hence, X will
2010.14265_F00360	14	■ 0.633	$\operatorname{MB}(T) \backslash \{X\}$	$\operatorname{MB}(T) \backslash \{X\}$	dependent on T conditioned on a subset of $\operatorname{MB}(T) \backslash \{X\}$ and
2010.14265_F00361	14	■ 0.381	$X \perp_P T \mid \boldsymbol{S} \backslash \{X\}$	$X \perp_P T \mid \boldsymbol{S} \backslash \{X\}$	thus $X \not\perp_P T \mid \boldsymbol{S} \backslash \{X\}$. In particular, this subset consists of
2010.14265_F00362	14	■ 0.980	$W \notin \operatorname{MB}(T)$	$W \notin \operatorname{MB}(T)$	Last, assume $W \notin \operatorname{MB}(T)$, but $W \in S$ after the grow
2010.14265_F00363	14	■ 0.980	$W \in S$	$W \in S$	Last, assume $W \notin \operatorname{MB}(T)$, but $W \in S$ after the grow
2010.14265_F00364	14	■ 0.958	$\boldsymbol{S} \backslash \{W\}$	$\boldsymbol{S} \backslash \{W\}$	phase. Further, we can write $\boldsymbol{S} \backslash \{W\}$ as $\operatorname{MB}(T) \cup \boldsymbol{Q}$, where
2010.14265_F00365	14	■ 0.958	$\operatorname{MB}(T) \cup \boldsymbol{Q}$	$\operatorname{MB}(T) \cup \boldsymbol{Q}$	phase. Further, we can write $\boldsymbol{S} \backslash \{W\}$ as $\operatorname{MB}(T) \cup \boldsymbol{Q}$, where
2010.14265_F00366	14	■ 0.975	$\boldsymbol{Q}$	$\boldsymbol{Q}$	$\boldsymbol{Q}$ contains all elements from $\boldsymbol{S} \backslash \{W\}$ that are not in $\operatorname{MB}(T)$.
2010.14265_F00367	14	■ 1.000	$T \perp_G \{W\} \cup \boldsymbol{Q} \mid \operatorname{MB}(T)$	$T \perp_G \{W\} \cup \boldsymbol{Q} \mid \operatorname{MB}(T)$	Then, $T \perp_G \{W\} \cup \boldsymbol{Q} \mid \operatorname{MB}(T)$ and thus by weak union,
2010.14265_F00368	14	■ 0.990	$T \perp_G W \mid \operatorname{MB}(T) \cup \boldsymbol{Q}$	$T \perp_G W \mid \operatorname{MB}(T) \cup \boldsymbol{Q}$	$T \perp_G W \mid \operatorname{MB}(T) \cup \boldsymbol{Q}$, which implies $T \perp_P W \mid \boldsymbol{S} \backslash \{W\}$ (by
2010.14265_F00369	14	■ 0.990	$T \perp_P W \mid \boldsymbol{S} \backslash \{W\}$	$T \perp_P W \mid \boldsymbol{S} \backslash \{W\}$	$T \perp_G W \mid \operatorname{MB}(T) \cup \boldsymbol{Q}$, which implies $T \perp_P W \mid \boldsymbol{S} \backslash \{W\}$ (by